A2	# Preprocessing and Visualization	**Artificial Intelligence** **AIN2IL (MTD.ba)** **SS2026**
	Submission via Moodle	**30 Points**

Exercise 1 (30 points) Load the breast-cancer-wisconsin data set from the UCI Machine Learning repository (file `wdbc.data`) and analyze it using a notebook (you may use the notebook `Preprocessing_Visualization.ipynb` as a template). First load and pre-process the data properly (in particular, determine meaningful feature names; transform the data frame into standard format used for scikit-learn data sets; information can be found in the file `wdbc.names`). Then perform the following analyses and visualizations:

1. Bar chart and pie chart for class column
2. For all numerical columns:
   - Compute minima, maxima, mean, and standard deviations
   - Histogram and box plot
   - Histogram (overlaid) and box plot per class
3. Create ten scatter plots using pairs of numerical columns with samples labeled by class (choose primarily those features that seemed most promising from the labeled histograms)
4. Perform PCA and visualize a scatter plot of the first two components labeled by class

Discuss your results: from the results that you have obtained, would you infer that the classification of cancer malignancy is possible? Hand in your notebook and a PDF or HTML dump.